Stance on a claim

The design

Every passage in the corpus is read by a language model and labeled for its stance toward one claim. Nothing is sampled, so there is no sampling error and no weights. What is left is the variation of the shares themselves and the error in the labels, and only the first of those is in the intervals below.

1. **Coding.** Each passage goes to the model alone, in its own request, at temperature 0, without its date or its document. `st$workers` requests run at once; identical passages are sent once and the label copied to each. The prompt is built from the scheme in `st$scheme` (Favor, neutral, oppose). The model first decides whether the passage addresses the claim at all (otherwise *irrelevant*), then which label fits.
2. **Estimates.** Salience plus one quantity per label, because a share of all passages confounds two different things:

$$P(\text{favor}) = P(\text{addresses}) \times P(\text{favor} \mid \text{addresses})$$

- **Salience** is the share of *all* passages that address the claim, that is $1 - P(\text{irrelevant})$. It says whether the claim is live in the corpus.
- **The labels** (favor, neutral, oppose for the default) are shares of the passages that *do* address it. They say which way those writers lean.

Reported together, the two separate a rise in agreement from a rise in attention; a single combined share cannot tell them apart. Multiplying them back together recovers a share of all passages. The cost is that the direction estimates rest on a smaller base that moves over time, so a period can hold plenty of passages and still hold too few that address the claim. Each basis therefore gets its own period set, and the report says what was dropped.

Each quantity is modeled as a smooth function of time (a logistic regression on a natural spline, `df` = 2), and passages from one document are correlated, so the 95 percent intervals are clustered by document.

The rules that decide whether there is a trend at all. A period needs at least 30 passages to appear in one, and a trend needs at least 4 such periods; otherwise time is dropped and the estimate is static for the whole corpus. Periods under the threshold are left out of the trend but stay in the static estimate. These thresholds are a judgment, not a standard: 30 is about where a share stops moving on one or two passages, and 4 is the fewest periods on which a one-knot spline says anything a straight line would not.

Passage length. Only passages between 5 and 250 words are coded. Below the floor there is not enough text to hold a position; above the ceiling the text is a document rather than a passage, holds several positions at once, and may run past the model's context. These bounds are a guard

against the wrong unit of text. They are much wider than what has been tried, which is paragraphs of roughly 50 to 80 tokens.

The app runs these same functions on a file a user uploads, so a claim submitted there is treated exactly as the one below.

Scheme: Favor, neutral, oppose

Claim: The United States should take an active role in the affairs of other nations.

Model: gemma4:12b-it-qat

The corpus

951 rows in comments_periods.csv; 877 within the word bounds, from 60 document(s).

Rows left uncoded.

why	rows
over 250 words	74

Coding

Labels are saved every 200 passages, so an interrupted run resumes where it stopped. Changing the claim, the scheme, or the model refuses the saved labels and names the file to delete; changing the corpus file does the same.

Labels for 877 passages from 60 document(s).

label	n	share
favor	159	0.181
irrelevant	658	0.750
neutral	30	0.034
oppose	30	0.034

What the labels look like

Three passages drawn at random for each label. If these do not match what the label should mean, fix the claim before reading any number.

FAVOR | row 764 - text: These will be years when Americans have restored their confidence and tradition of progress; when our values of faith, family, work, and neighborhood were restated for a modern age; when our economy was finally freed from government's grip; when we made sincere efforts at meaningful arms reduction, rebuilding our defenses, our economy, and devel…

FAVOR | row 480 - text: That all nations are equally interested in the peace of the world and in the political stability of free peoples, and equally responsible for their maintenance; that the

essential principle of peace is the actual equality of nations in all matters of right or privilege; that peace cannot securely or justly rest upon an armed balance of power; th…

FAVOR | row 511 - text: My countrymen, no one can contemplate current conditions without finding much that is satisfying and still more that is encouraging. Our own country is leading the world in the general readjustment to the results of the great conflict. Many of its burdens will bear heavily upon us for years, and the secondary and indirect effects we must expect …

NEUTRAL | row 495 - text: Amid it all we have riveted the gaze of all civilization to the unselfishness and the righteousness of representative democracy, where our freedom never has made offensive warfare, never has sought territorial aggrandizement through force, never has turned to the arbitrament of arms until reason has been exhausted. When the Governments of the ea…

NEUTRAL | row 733 - text: The world is still engaged in a massive armaments race designed to ensure continuing equivalent strength among potential adversaries. We pledge perseverance and wisdom in our efforts to limit the world's armaments to those necessary for each nation's own domestic safety. And we will move this year a step toward ultimate goal - the elimination of…

NEUTRAL | row 269 - text: It will be my endeavor to execute all laws in good faith, to collect all revenues assessed, and to have them properly accounted for and economically disbursed. I will to the best of my ability appoint to office those only who will carry out this design. In regard to foreign policy, I would deal with nations as equitable law requires individuals …

OPPOSE | row 141 - text: Our course of foreign policy has been so uniform and intelligible as to constitute a rule of Executive conduct which leaves little to my discretion, unless, indeed, I were willing to run counter to the lights of experience and the known opinions of my constituents. We sedulously cultivate the friendship of all nations as the conditions most comp…

OPPOSE | row 486 - text: The recorded progress of our Republic, materially and spiritually in itself proves the wisdom of the inherited policy of noninvolvement in Old World affairs. Confident of our ability to work out our own destiny, and jealously guarding our right to do so, we seek no part in directing the destinies of the Old World. We do not mean to be entangled….

OPPOSE | row 159 - text: Our Confederacy, fellow-citizens, can only be preserved by the same forbearance. Our citizens must be content with the exercise of the powers with which the Constitution clothes them. The attempt of those of one State to control the domestic institutions of another can only result in feelings of distrust and jealousy, the certain harbingers of d…

Results

The whole corpus, with 95 percent intervals clustered by document. Salience is out of all passages; the labels are out of the passages that address the claim.

quantity	basis	passages	share	lo	hi
salience	all passages	877	0.250	0.198	0.310
favor	passages that address the claim	219	0.726	0.622	0.810
neutral	passages that address the claim	219	0.137	0.089	0.205
oppose	passages that address the claim	219	0.137	0.084	0.216

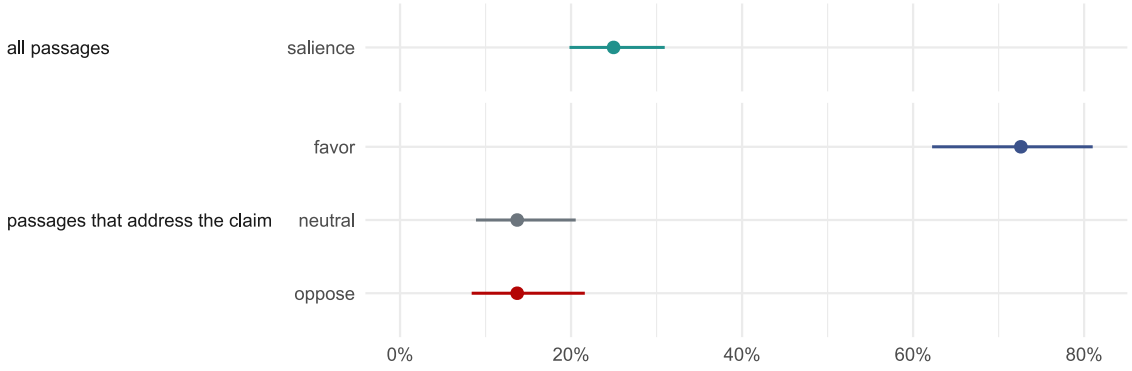


Figure 1: The whole corpus. Intervals are 95 percent and do not include error in the model's labels.

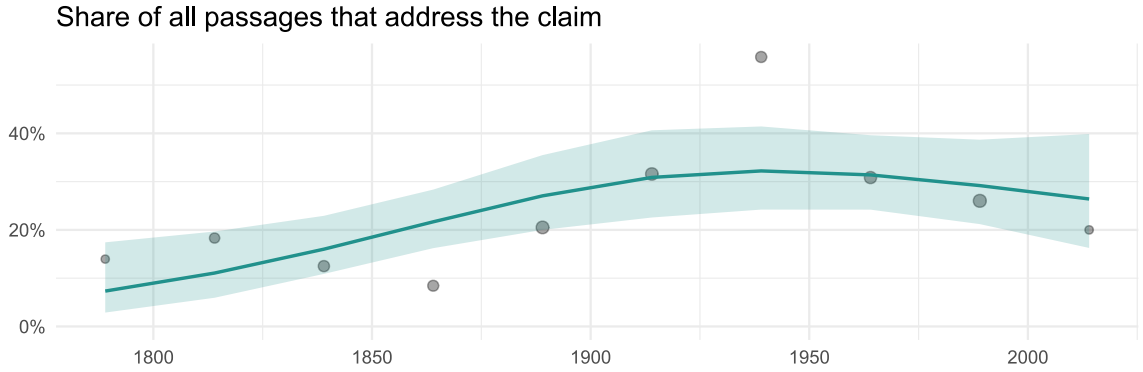


Figure 2: Salience: the share of all passages that address the claim. Points are the observed share in each period, sized by the passages behind it. Intervals are 95 percent, clustered by document, and do not include error in the model's labels.

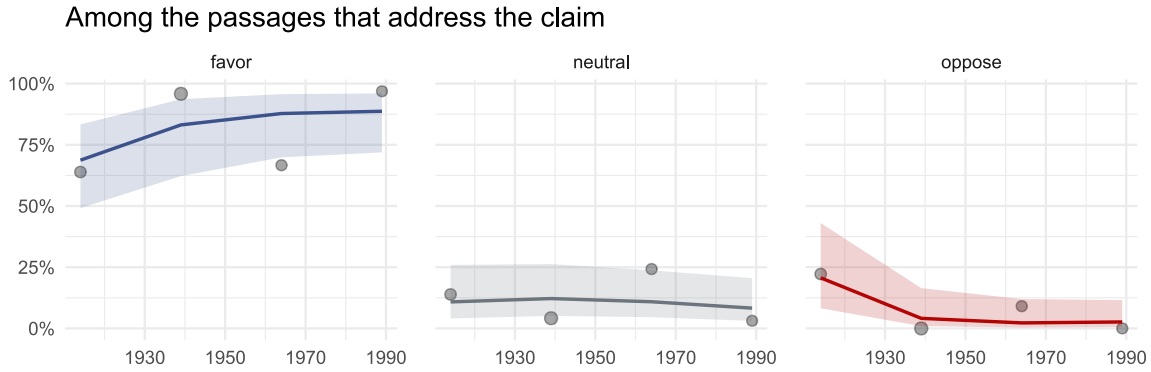


Figure 3: The labels among the passages that address the claim, on one scale because they are shares of the same passages and add to 100 percent. Points and intervals as in the salience figure.

Checking the labels

The intervals above treat the labels as correct. They are not, and nothing in the pipeline measures how often they are wrong. The only way to find out is to code some passages by hand and compare.

Draw 400 at random, code them without seeing the model's label, and read three things: overall agreement, Cohen's kappa, and the confusion matrix. 400 puts agreement within about 5 percentage points, which is enough to tell a usable coder from a broken one. It is not enough to say much about a label the model rarely uses: a label on 5 percent of passages turns up about 20 times, and 20 cases carry roughly 18 points of uncertainty, so read the per-label rows with their counts beside them.

Two cautions. One coder is a weak standard, so for a decision that matters have a second person code a subset and compare the coders first. And agreement alone hides direction: a model that pushes everything into one label can still agree often when that label is common, which shows up in the confusion matrix as a filled column.

Wrote stance_handcode_world_role.csv: 400 passages with a blank `hand_label` column and no model label. Code it, put it in stance/inputs, and name it in `st$hand_file` to see the comparison here.

No hand codes configured, so **the accuracy of these labels is unmeasured.** Say so wherever the estimates are reported.

Before using these numbers

- The labels come from one model. Their error is not in the intervals.
- Wording moves the answer. Re-run with a paraphrase of the claim and compare before relying on a result.
- Every share is a share of passages, so a long document contributes more than a short one. The document column makes the intervals account for that; it does not make the shares equal per document.

- The corpus is not a sample of anything wider. The estimates describe this corpus.
- Gaps in time are never drawn as observed: a period with no passages, or too few, is absent from the curve rather than interpolated across.